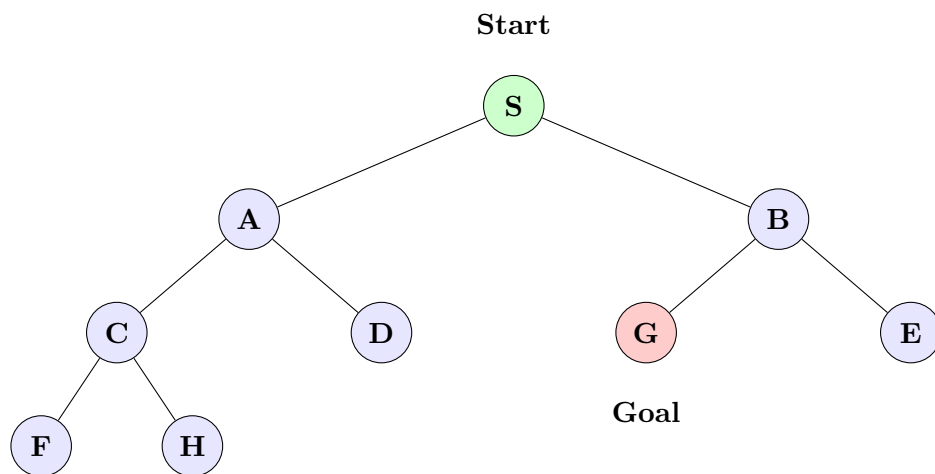


CS471 – Lesson 5 Solutions Key

Uninformed Search (DFS, BFS, UCS)

Guided In-Class Worksheet Solutions

Scenario 1: A rescue drone is navigating a decision tree to find a lost squad (Goal 'G').



Part 1: Search Space Parameters & Uninformed Trees

1. Branching Factor (b):

Solution

Answer: The Branching Factor (b) is **2**. (Nodes S, A, C, and B all have 2 children).

2. Maximum Depth (m):

Solution

Answer: The Maximum Depth (m) is **3**. (Path $S \rightarrow A \rightarrow C \rightarrow F$ is 3 steps).

3. Depth of Shallowest Solution (s):

Solution

Answer: The Shallowest Solution Depth (s) is **2**. (Path $S \rightarrow B \rightarrow G$).

4. Depth-First Search (DFS) Expansion Order:

Solution

Answer: S, A, C, F, H, D, B, G.

(DFS dives down the left-most path first. It goes $S \rightarrow A \rightarrow C \rightarrow F$ (dead end). Backs up to $C \rightarrow H$. Backs up to $A \rightarrow D$. Backs up to $S \rightarrow B \rightarrow G$.)

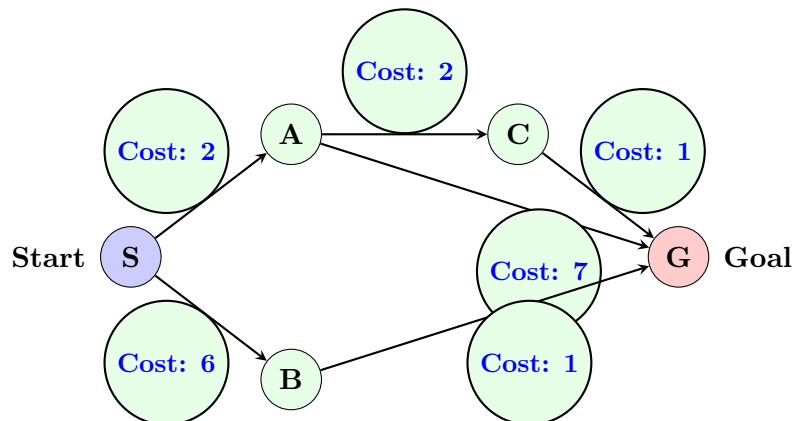
5. Breadth-First Search (BFS) Expansion Order:

Solution

Answer: S, A, B, C, D, G.

(BFS explores tier-by-tier. Tier 0: S. Tier 1: A, B. Tier 2: C, D, G.)

Scenario 2: A convoy must route through a map where roads vary in terrain. The convoy must find the absolutely optimal (cheapest) route to the Goal.



Part 2: Uniform Cost Search (UCS)

6. Trace the Priority Queue:

Solution

Step 1: Start node added. Queue: $[S(0)]$

Step 2: Expand S. Add A and B. Queue: $[A(2), B(6)]$

Step 3: Expand A. Add C and G. Queue: $[C(4), B(6), G(9)]$

Step 4: Expand C. Add G (Cost $4+1=5$). Update G's cost. Queue: $[G(5), B(6)]$

Step 5: Expand G. It is the goal!

Expansion Order Answer: S, A, C, G.

7. Final Path & Optimality:

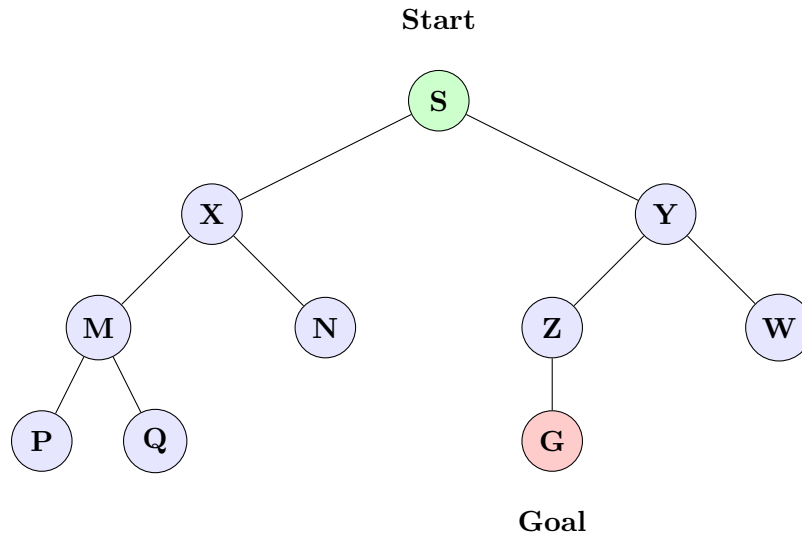
Solution

Final Path: $S \rightarrow A \rightarrow C \rightarrow G$

Total Cost: 5

Student Practice Solutions

Scenario 3: A surveillance drone must navigate the following communication tree to locate a specific signal broadcast (Goal 'G').



Part 3: Uninformed Trees Practice

8. Depth-First Search (DFS) Trace:

Solution

Answer: S, X, M, P, Q, N, Y, Z, G.

(DFS relentlessly pursues the left-most branch down to its leaves before ever checking the right branch (Y)).

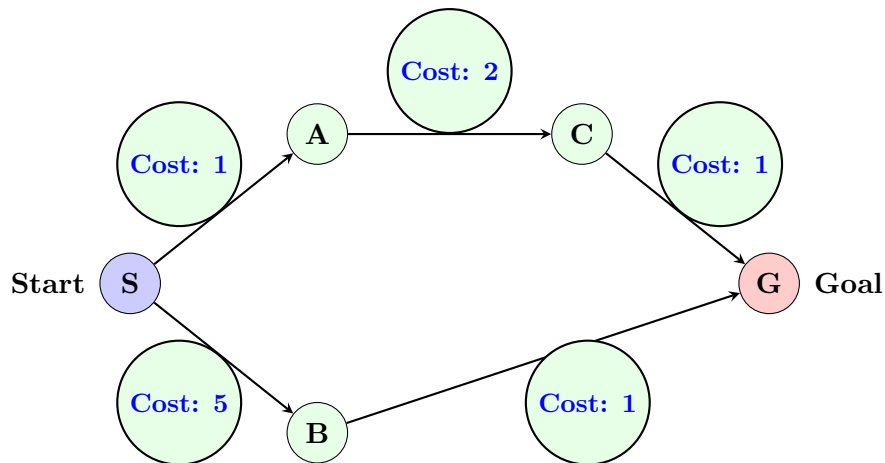
9. Breadth-First Search (BFS) Trace:

Solution

Answer: S, X, Y, M, N, Z, W, P, Q, G.

(BFS checks Tier 1 (X, Y), then fully checks Tier 2 (M, N, Z, W) before finally moving to Tier 3 where the Goal resides).

Scenario 4: A ground extraction team must navigate a hostile urban grid. Moving down main avenues is fast (Cost 1), but moving through alleyways is dangerous and slow (Cost 5).



Part 4: Uniform Cost Search (UCS) Practice

10. Trace the Queue:

Solution

Step 1: Start node added. *Queue:* $[S(0)]$
Step 2: Expand S. Add A and B. *Queue:* $[A(1), B(5)]$
Step 3: Expand A. Add C. *Queue:* $[C(3), B(5)]$
Step 4: Expand C. Add G. *Queue:* $[G(4), B(5)]$
Step 5: Expand G. It is the goal!
Expansion Order Answer: S, A, C, G.

11. Final Path & Breadth-First Flaw:

Solution

Final Path: $S \rightarrow A \rightarrow C \rightarrow G$
Total Cost: 4

Why BFS fails: BFS evaluates strictly by the number of steps (depth). The path $S \rightarrow B \rightarrow G$ is only 2 steps, so BFS would blindly select it. However, the true total cost of that path is 6 ($5+1$), which is worse than the 3-step path (cost 4). BFS is blind to varying edge weights.